



Student Name: Arsam Ali

Roll No: 238-0038

Program: AI

Examination: Final

Semester: Fall 2023

Total Marks/Weightage: 55

Time Allowed: 02 hours 30 minutes

Instructor: Fariba Laiq

Course: Programming Fundamentals

Date: 11th Dec 2023

NOTE:

- Attempt all questions.
- Don't rush, carefully understand the problem and then go for solution.
- Understanding the question is the part of Exam.
- An individual may be assigned a straight-forward 0 if the submitted assessed task is copied/cheated from another individual/internet/found using internet or chatGPT.
- Create a folder named as Roll number, name and section, copy all the files (Q1.c, Q2.c,), in that folder and submit the folder.

Smart Home Temperature Controller

Weightage [20]

Develop a program for a smart home system that controls the heating and cooling. The program should take the input temperature from the user and adjust the heating or cooling to maintain a set target temperature. After that, the program should ask if the user wants to enter the temperature again, and repeat the same steps.

- If the temperature is above 24 degrees centigrade, activate cooling. $temp > 24 \rightarrow cool$
- If below 22 degrees centigrade, activate heating. $\rightarrow temp < 22 \rightarrow heat$
- If between 22 and 24 (inclusive), it is considered as comfort mode, so no need to activate anything. $22 - 24 - no\ change$
- Must add validity checks for the temperature. In case of invalid input, the program should ask for an input again. $invalid \rightarrow 30$
- Do this task using functions. Make a function named `tempControl()` that takes the current temperature as an argument, and prints the mode: heating mode, cooling mode, or comfort mode.

1

input $\rightarrow$ temp $\rightarrow$ adjust heat/cool. $\rightarrow$ Ask user want to add temp.

Sample Output:

Enter current temperature in Centigrade: 28

Cooling mode activated.

Do you want to enter another temperature? (y/n): y

Enter current temperature in Centigrade: 21

Heating mode activated.

Do you want to enter another temperature? (y/n): y

Enter current temperature in Centigrade: 500

Invalid input. Please enter a temperature between -10 and 50 degrees.

Enter current temperature in Centigrade: 24

Comfort mode, no action needed.

Do you want to enter another temperature? (y/n): n

Fitness Tracker Data Analysis

Weightage [20]

Create a program for a fitness tracker that records daily step counts for a week. The user enters their step count for each day of the week into a 1D array. Design functions to calculate the total steps, average daily steps, and to identify the day with the highest step count. All of these functions should take the array as an argument. Include an option to set a weekly goal and a function to check if the goal was met, or not and display messages to user accordingly. Make sure to include the validity checks for the step count.

Sample Output:

Enter steps for day 1: 300

Enter steps for day 2: 500

Enter steps for day 3: 400

Enter steps for day 4: 700

Enter steps for day 5: 1000

total steps
average steps
max step day

Enter steps for day 6: 2000

Enter steps for day 7: 3000

Set your weekly step goal: 5000

Total steps: 7900

Average daily steps: 1128.57

Highest step count was on day: 7

Great job! You've met your goal!

Sum of Boundary Elements

Weightage [15]

Write a program that takes a 2D array as an input, pass it to a function, and prints the boundary elements, and sum of boundary elements of the array. Note that the dimension of the 2D array will be entered by the user.

You must take the dimensions and the elements of the array from user and your program must work correct for any dimensions of the array.

For example:

If the entered matrix was

1 2 3

4 5 6

7 8 9

Boundary Matrix:

1 2 3

4 6

7 8 9

Sum of boundary is 40

3 ← 3

2 3 4
5 6 7 3

8 9 10 0 0

(0, 0) (0, 1) (0, 2)